COURSE 102

Income Approach to Valuation

Solutions Set

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Chapter 1 | Assessment and Appraisal Theory

REVIEW QUESTIONS - Chapter 1 SOLUTION

1. One of three approaches to value in which the appraiser derives a value indication by converting anticipated benefits through ownership of income-producing property is the **income** approach.

2. Real estate competes with other investments for the investor's dollar. As an investor analyzes various opportunities, what will he or she consider?

   - **How much will it cost?**
   - **How much will I get back?**
   - **When will I get it back?**
   - **What are the risks?**
   - **What is the return of a real estate investment compared to other investments of similar risks?**

3. The economic principle of **anticipation** states that value is created by the expectation of benefits to be derived in the future.

4. The economic principle of **substitution** states that a property's maximum value tends to be set by the lowest cost or price at which another property of equivalent utility can be acquired.

5. Competition is created by the potential for profits. However, competition among sellers may lead to a/an **oversupply**, which reduces prices and profits. Competition among buyers may lead to **shortage**, which increases prices and profits to sellers.

6. The **recapture** rate reflects the return of the investment in the wasting asset.

7. List nine factors influencing investor decisions.
   - **Safety of the investment**
   - **Liquidity of the investment**
   - **Size of the investment**
   - **Use as collateral**
   - **Time**
   - **Management**
   - **Appreciation**
   - **Income tax advantages**
   - **Leverage**
8. **Positive** leverage is achieved when funds are invested in property, which has a higher rate of return than the cost of borrowed funds.
9. The four most common methods of financing real estate are:
   - **Mortgage**
   - **Trust deed**
   - **Cash**
   - **Land contract (contract for deed)**
10. A mortgage on personal property is termed a **chattel** mortgage.
11. A **junior** mortgage is also called a second mortgage.
12. List four types of mortgages classified according to repayment provisions.
    - **Amortized**
    - **Straight**
    - **Reverse**
    - **Partially amortized**
13. A payment on the balance due of a note at the end of the loan term that is in excess of the regular payment amounts is called a **balloon** payment.
14. A **package** mortgage covers real estate as well as personal property included with the real estate.

15. The four variables in real estate financing that affect the mortgage payment are:
    - **Amount borrowed**
    - **Interest rate**
    - **Term**
    - **Frequency of payment**
16. The **overall yield** rate (or discount rate) reflects the return on the investment.
17. The summation concept is a theoretical procedure in developing the discount rate for a real estate investment and is comprised of the following four parts.
    - **Safe rate**
    - **Risk rate**
    - **Rate for non-liquidity**
    - **Rate for management**
18. The **effective tax** rate reflects the relationship between the real estate taxes and the value of the property.
19. To obtain value using the IRV formula, one must **divide** the income by the rate.
20. To obtain value using the VIF formula, one must **multiply** the income by the factor.

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Chapter 2 | Development of the Net Operating Income Estimate

Exercise 2-1: Lease Analysis - Solution

Bob recently leased a 1,500 square foot space at a local shopping center. He signed a five-year lease, which requires him to pay $50,000 in rent over the life of the lease. Bob's lease is typical for new rental space in the area. Jeff, an associate of Bob, leased retail space at the same shopping center two years ago and has three years left on his current lease agreement. Jeff has 2,000 square feet of leased area, which he rented for $4.00 a square foot two years ago. His lease calls for annual rent adjustment of 2.5% per year. He also pays the landlord an additional three percent of all his gross sales above $500,000 per year. Last year Jeff had $800,000 in gross sales.

What is the annual rent per square foot that Bob pays?

$50,000 / 5 = $10,000 / 1,500 square feet = $6.67 / sf

What type of rent is it?

Market Rent

What type of rent does Jeff pay?

Percentage Rent

How much rent did Jeff have to pay the landlord last year?

$4.00 × 1.025 = $4.10

Base Rent: 2,000 x $4.10 ($4.00 x 1.025) = $8,200

Overage Rent: $800,000 - $500,000 = $300,000 x .03 = $9,000

Total Rent: $8,200 + $9,000 = $17,200

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Exercise 2-2: Lease Analysis - Solution

Annual Rent of subject property

Minimum Base Rent: 2,500 SF × $5.00 - $12,500

Overage Rent: $475,000 – $100,000 = $375,000 × 0.03 = $11,250

Total Annual Rent: $12,500 + $11,250 = $23,750

Type of lease utilized: Percentage Lease

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Exercise 2-3: Unit of Comparison-Rent per Square Foot - Solution

ANNUAL RENT PER SQUARE FOOT (GROSS LEASABLE AREA)		
Annual Rent	$380,000	
Square Feet of Gross Leasable Area: 16,000 sq. ft.	÷ 16,000	Sq. Ft.
Annual Rent per Square Foot (GLA) ($380,000 ÷ 16,000) =	**$23.75**	

ANNUAL RENT PER SQUARE FOOT (NET LEASABLE AREA)		
Calculate Net Leasable Area		
Square Feet of Gross Leasable Area: 16,000 sq. ft.	16,000	Sq. Ft.
Less Common Space: 3,000 sq. ft.	- 3,000	Sq. Ft.
Total Net Leasable Area	**13,000**	**Sq. Ft.**
Calculate NLA Rate		
Annual Rent	$380,000	
Net Leasable Area	÷ 13,000	Sq. Ft.
Annual Rent per Sq. Ft. (NLA) ($380,000 ÷ 13,000) =	**$29.23**	

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Exercise 2-4: Unit of Comparison-Apartment Unit vs. SQFT - Solution

One-Bedroom Apartments				Two-Bedroom Apartments		
Rental No.	**Sq. Ft.**	**$/Apt. Unit**	**$/Sq. Ft.**	**Sq. Ft.**	**$/Apt. Unit**	**$/Sq. Ft.**
1	575	$820	$1.43	810	$890	$1.10
2	550	740	$1.35	840	940	$1.12
3	580	840	$1.45	850	1,000	$1.18
4	535	750	$1.40	775	870	$1.12
5	525	720	$1.37	780	860	$1.10
Range	Low	$720	$1.35	Low	$860	$1.10
	High	$840	$1.45	High	$1,000	$1.18
Spread		$120	$0.10		$140	$0.08
÷ Low		$720	$1.35		$860	$1.10
% difference		**16.67%**	**7.41%**		**16.28%**	**7.27%**

Analysis:

The rental for one-bedroom apartments ranges from $720 to $840 per month and that of the two-bedroom apartments from $860 to $1,000 per month. The rate per square foot for one-bedroom units the range is from $1.35 to $1.45 per square foot and for two-bedroom apartments, $1.10 to $1.18.

Recommendation:

This substantial difference of income between the properties indicates that per apartment unit would not be as accurate a unit of comparison compared to the square foot analysis. Of the two units of comparison considered, rent per square foot is the most consistent throughout the properties. Since the rent per square foot is so consistent, the recommended unit of comparison is the rent per square foot.

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Exercise 2-5: Development of Market Rental Estimate - Solution

One-Bedroom Units		**Monthly Unit Size**	**Monthly Rent**	**Monthly Calculation**	**Monthly Rent/SQFT**
Alexan Palm Valley		760	$840	$840 ÷ 760 =	$1.11
Alexan Paradise Lane		760	$850	$850 ÷ 760 =	$1.12
Aventura Apartments		850	$925	$925 ÷ 850 =	$1.09
Rock Canyon		850	$895	$895 ÷ 850 =	$1.05
Sage Stone at Arrowhead		700	$770	$770 ÷ 700 =	$1.10
	Average	**784**	**$856**		**$1.09**
	Median	**760**	**$850**		**$1.10**
		High	$925		$1.12
		-Low	$770		$1.05
		Spread	**$155**		**$0.07**
		÷Low	$770		$1.05
Percentage Spread			**20%**		**6%**

Two-Bedroom Units		**Monthly Unit Size**	**Monthly Rent**	**Monthly Calculation**	**Monthly Rent/SQFT**
Alexan Palm Valley		1,125	$1,080	$1,080 ÷ 1,125 =	$0.96
Alexan Paradise Lane		1,250	$1,320	$1,320 ÷ 1,250 =	$1.06
Aventura Apartments		1,250	$1,200	$1,200 ÷ 1,250 =	$0.96
Rock Canyon		1,025	$985	$985 ÷ 1,025 =	$0.96
Sage Stone at Arrowhead		1,050	$990	$990 ÷ 1,050 =	$0.94
	Average	**1,140**	**$1,115**		**$0.98**
	Median	**1,125**	**$1,080**		**$0.96**
		High	$1,320		$1.06
		-Low	$985		$0.94
		Spread	**$335**		**$0.11**
		÷Low	$985		$0.94
Percentage Spread			**34%**		**12%**

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Rent per square foot has a lower percentage spread than the rent per unit indications and is the appropriate unit of comparison to apply in the analysis.

One-Bedroom Units	**Features (Elements of Comparison)**						
	# Baths	**SQFT**	**Covered Parking**	**Age (Years)**	**# Pools**	**# Club-houses**	**# of Similar Features**
Subject	1	850	Yes	5	2	2	N/A
Alexan Palm Valley	1	760	No	1	1	1	1
Alexan Paradise Lane	1	760	Yes	0	1	1	2
Aventura Apartments	1	850	Yes	5	2	2	6
Rock Canyon	1	850	No	5	2	1	4
Sage Stone at Arrowhead	1	700	Yes	5	2	1	4

Two-Bedroom Units	**Features (Elements of Comparison)**						
	# Baths	**SQFT**	**Covered Parking**	**Age (Years)**	**# Pools**	**# Club-houses**	**# of Similar Features**
Subject	1.5	1,250	Yes	5	2	2	N/A
Alexan Palm Valley	2	1,125	No	1	1	1	1
Alexan Paradise Lane	2	1,250	Yes	0	1	1	2
Aventura Apartments	1.5	1,250	Yes	5	2	2	6
Rock Canyon	2	1,075	No	5	2	1	4
Sage Stone at Arrowhead	1.5	1,050	Yes	5	2	1	4

Aventura Apartments is most similar to the subject in terms of having the greatest number of similar features as the subject, with a one-bedroom rent of $1.09 per square foot and a two-bedroom rent of $0.96 per square foot. The average for all five one-bedroom rents is $0.96 per square foot and the median is $1.09 per square foot. The average for all five two-bedroom rents is $0.96 per square foot and the median is $0.96 per square foot. The rents for Aventura Apartments are a good indication for the subject property, reflect the mid-points of the data sets, and are appropriate to weight in the concluded market rent for the subject.

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The total annual rent estimate for the subject is as follows:

Villas at Camelback Crossing							
Floor Plan	**SQFT**	x	**Market Rent**	x	**# of Units**	=	**Rent**
One-Bedroom	850	x	$1.09	x	300	=	$277,950
Two-Bedroom	1,250	x	$0.96	x	200	=	$240,000
Total Monthly Market Rent Estimate							**$517,950**
							x 12
Total Annual Market Rent Estimate							**$6,215,400**

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Exercise 2-6: Developing a Potential Gross Income Estimate - Solution

One-Bedroom Units					
Level	**Base Rent**	**Adjustment**	**Monthly Rent**	**Number**	**Gross Rent***
First	$ 700	1.00	$ 700	4	$ 2,800
Second	700	1.25	875	6	5,250

***All rents are monthly**

Two-Bedroom					
Level	**Base Rent**	**Adjustment**	**Monthly Rent**	**Number**	**Gross Rent***
First	$ 800	1.00	$ 800	2	$ 1,600
Second	800	1.25	1,000	4	4,000
Third	800	1.25	1,000	4	4,000

***All rents are monthly**

Three-Bedroom					
Level	**Base Rent**	**Adjustment**	**Monthly Rent**	**Number**	**Gross Rent***
First	$ 900	1.00	$ 900	2	$ 1,800
Third	900	1.25	1,125	4	4,500

***All rents are monthly** **Total $23,950**

$23,950 × 12 (months) = $287,400 - Potential Gross Income (PGI)

Note: The fact the janitor and the owners live in the apartment does not affect the estimate of PGI. This is a management decision and as the appraiser, you must look at the PGI for the apartment as the market rent at 100% occupancy.

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Exercise 2-7: Income & Expense Statement - Vacancy Rate - Solution

You are reconstructing an income and expense statement for an apartment complex containing 72 units. You have found four apartment complexes that have similar characteristics and amenities, located in the same neighborhood as your subject.

Complex	Units	Occupied	Vacant	Vacant ÷ Total Units	Vacancy Rate
Briargate	80	76	4	4 ÷ 80	0.050
Creekside	60	57	3	3 ÷ 60	0.050
Beachfront	78	74	4	4 ÷ 78	0.051
Lakeside	62	59	3	3 ÷ 62	0.048

The indicated vacancy rate for your reconstructed income and expense statement would be 0.05 or 5%.

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Exercise 2-8: Income & Expense Statement - Collection Loss Rate - Solution

You are reconstructing an income and expense statement for an apartment complex containing 120 units and need to determine a rate for collection loss. You have found four apartment complexes with rental information, similar characteristics and amenities, located in the same neighborhood as your subject.

Complex	Billable	Collected	Collection Loss	Loss ÷ Billable	Rate
Peachtree	$984,900	$965,100	$ 19,800	$19,800 ÷ $984,900	0.0201
Applewood	1,094,400	1,072,600	21,800	21,800 ÷ 1,094,400	0.0199
Elderberry	1,056,000	1,035,200	20,800	20,800 ÷ 1,056,000	0.0197
Brentwood	1,130,800	1,107,900	22,900	22,900 ÷ 1,130,800	0.0203

The indicated rate for collection loss in your reconstructed income and expense statement would be 0.02 or 2%.

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Exercise 2-9: Income & Expense Statement - Vacancy & Collection Loss Rate - Solution

Complex	Units	Occupied	Vacant	Vacant ÷ Total Units	Vacancy Rate
A	210	199	11	11 ÷ 210	0.0524
B	180	171	9	9 ÷ 180	0.0500
C	196	186	10	10 ÷ 196	0.0510
D	200	190	10	10 ÷ 200	0.0500
E	220	209	11	11 ÷ 220	0.0500

The indicated rate for vacancy would be 0.05 or 5%.

Complex	Billable	Collected	Collection Loss	Loss ÷ Billable	Rate
A	$1,925,300	$1,886,800	$38,500	$38,500 ÷ $1,925,300	0.0200
B	1,744,200	1,709,500	34,700	34,700 ÷ 1,744,200	0.0199
C	1,899,200	1,861,200	38,000	38,000 ÷ 1,899,200	0.0200
D	1,938,000	1,898,000	40,000	40,000 ÷ 1,938,000	0.0206
E	2,131,800	2,089,300	42,500	42,500 ÷ 2,131,800	0.0199

The indicated rate for collection loss would be 0.02 or 2%.

The rate for vacancy and collection loss would be the rate for vacancy combined with the rate for collection loss.

The rate for vacancy and collection loss would be 0.07 or 7%. This would be the rate used in the reconstructed income and expense statement.

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Exercise 2-10: Reconstruction of an Income Statement - Solution

	Expense Item	Use as Stated	Pro-Rate	Eliminate
A	Management fee	X		
B	Repairs	X		
C	Miscellaneous	X		
D	Utilities	X		
E	Interest on mortgage			X
F	Principal on mortgage			X
G	New roof		X	
H	Insurance fire (3-year policy)		X	
I	Insurance liability (1-year policy)	X		
J	Employee's health policy (1-year)	X		
K	Janitor's salary	X		
L	Painting exterior		X	
M	Purchase of 3 new refrigerators		X	
N	Purchase of 4 range/ovens		X	
O	Supplies	X		
P	Corporate income taxes			X
Q	Red Cross donation			X
R	Carpet replacement (4 units)		X	
S	Redecorate 7 apartment units		X	
T	Real estate taxes			X

Principal and Interest on mortgage are accounted for in the capitalization rate and thus should be eliminated as an operating expense.

Any appropriate expenses (such as a new roof, 3-year insurance policy, new ranges and refrigerators, carpet, or redecoration expenses) which last more than one year should be pro-rated.

Corporate taxes or donations are not expenses of the real estate but corporate expenses.

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Real estate taxes are appropriate expenses of the real estate, however because our appraisal is to be used as the basis for future taxes, we must eliminate property tax as an expense and add the effective tax rate to the overall capitalization rate. Real estate taxes must be eliminated because we do not know how much the real estate tax expense will actually be until we do the property tax appraisal. If we left the real estate tax in the expenses as a legitimate expense, we would have to pre-suppose the value before the appraisal in order to compute the tax.

Exercise 2-11: Reconstruction of an Income Statement - Solution

Income or Expense Items	Owner's Statement	Reconstructed Statement
Gross Income	**$ 365,000**	**$ 365,000**
Vacancy & Collection Loss	0	14,600
Effective Gross Income	**$ 365,000**	**$ 350,400**
Expenses:		
Management ($350,400 x 0.05)		$ 17,520
Employees' salaries	$ 34,022	34,022
Employees' benefits	2,160	2,160
Insurance	7,000	7,000
Natural gas	13,445	13,445
Painting & decorating (5 units) *	6,000	12,000
Repairs	8,425	8,425
Supplies	2,040	2,040
Electricity	4,850	4,850
Water	1,680	1,680
Real Estate Taxes	32,550	
Depreciation	56,000	
Interest on mortgage	120,650	
Legal & accounting fees	4,800	4,800
Principal on mortgage	19,100	
Miscellaneous expense	3,150	3,150
Total Expenses	**$ 315,872**	**$ 111,092**
Net Operating Income	**$ 49,128**	**$ 239,308**

*$6,000 ÷ 5 units = $1,200

$1,200 x 30 units at subject = $36,000

$36,000 ÷ 3-year economic life = $12,000

Operating Expense Ratio:

$111,092 / $350,400 = 0.317 or 32%

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Exercise 2-12: Reconstruction of an Income Statement - Solution

Income or Expense Items	Owner's Statement	Reconstructed Statement
Gross Income	**$ 697,500**	**$ 750,000**
Vacancy & Collection Loss	0	- 37,500
Miscellaneous Income (annual)	+ 2,750	+ 2,750
Effective Gross Income	**$ 700,250**	**$ 715,250**
Expenses:		
Management		$ 28,610
Interest on mortgage	168,238	
Insurance	4,500	4,500
Real Estate taxes	112,500	
Janitor's salary	36,000	36,000
Water	11,000	11,000
Electricity	16,000	16,000
Natural gas	18,000	18,000
Miscellaneous repairs	11,250	11,250
Supplies	3,375	3,375
Principal on mortgage	70,000	
Exterior repairs	31,250	31,250
Depreciation	72,000	
Building addition	120,000	
6 Refrigerators (1,750 x 50 ÷ 10)	10,500	8,750
8 Ranges (1,500 x 50 ÷ 10)	12,000	7,500
2 Water heaters(2,250 x 5 ÷ 5)	4,500	2,250
Roof cover (45,000 ÷ 12)		3,750
Floor cover (37,500 ÷ 5)		7,500
Redecorating (3,750 x 50 ÷ 5)		37,500
Total Expenses	**$ 701,113**	**$ 227,235**
Net Operating Income	**$ (863)**	**$ 488,015**

Calculation of Potential Gross Income is as follows:

Efficiency units	10 x $1,000 x 12 months	=	$120,000
One bedroom units	30 x $1,250 x 12 months	=	$450,000
Two bedroom units	10 x $1,500 x 12 months	=	$180,000
	Total Potential Gross Income	=	$750,000

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Expense explanation:

Items such as interest on the mortgage, principal on the mortgage, depreciation, and the new building addition, would not be considered either annual expenses or expenses necessary to maintain the property and would not be included in the reconstructed operating statement. Real estate taxes are included as a part of the capitalization rate and would not be deducted as an operating expense. Also, all short-lived items should be included in the reconstructed operating statement as replacement reserves.

The operating statement after reconstruction provides a positive net operating income with an expense ratio of 31.77%.

REVIEW QUESTIONS - Chapter 2 SOLUTION

1. In addition to sales and leases, **offers** and **listings** can be a valid source of market information, if care is exercised.
2. List five sources of rental income data:
   - **Commercial property managers**
   - **Tenants of comparable properties**
   - **Real estate brokers and salespeople**
   - **Sellers and purchasers of comparable properties**
   - **Public records**
3. Another name for market rent is **economic** rent.
4. A lease that provides for one or more extensions of the lease term in the original lease document at the option of the tenant is a **renewal** lease.
5. The fixed portion of the rent under the terms of a percentage lease is known as **minimum** **base rent**.
6. When contract rent exceeds economic rent, the difference is known as **excess** rent.
7. The rent that is over and above a guaranteed minimum base rent is referred to as **overage** rent.
8. A contract that requires a fixed minimum base rent and a variable rent based on volume of business, sales, productivity, or use of the property by the tenant is called a **percentage** lease.
9. A lease in which the landlord is required to pay all operating expenses associated with the real estate is known as a **gross** lease.

10. A lease in which the tenant is required to pay all operating expenses associated with the real estate is known as a **triple net** lease.

11. List four factors that must be considered when comparing rental properties.
    - **Date of lease**
    - **Location of property**
    - **Physical characteristics of property**
    - **Terms of lease**

12. List five common units of comparison when comparing rental properties.
    - **Rent per unit**
    - **Rent per room**
    - **Rent per space**
    - **Rent per square foot**
    - **Gross income multiplier**

13. The total leasable area, including common areas is known as **gross leasable area**.

14. The **effective gross income** is the potential gross income, less vacancy and collection loss, plus appropriate miscellaneous income.

15. Income from vending machines and coin-operated laundries is considered examples of **miscellaneous** income.

16. The first step in reconstructing the income and expense statement is to determine the **potential gross income**.

17. Which of the following would be considered a proper expense in the reconstructed operating expense statement?
    a. Salaries
    b. Management
    c. Utilities
    d. <u>All of the above</u>
18. Expense amounts for short-lived items such as carpet, drapes, and water heaters are considered **<u>reserves for replacement</u>**.
19. When reconstructing an income/expense statement, depreciation and debt service would be considered a/an **<u>improper</u>** expense.
20. The income remaining after developing effective gross income and allowing for operating expenses and reserves for replacement is known as **<u>net operating income</u>**.
21. **<u>Rent Concessions</u>** take the form of either free rent or extra tenant improvement allowances.
22. **<u>Leased fee value</u>** tends to be less than fee simple value because leases are typically set at market level when written, but usually fall below market levels as time passes.
23. **<u>Class B</u>** buildings are generally a little older but still have good quality management and tenants.
24. **<u>Effective</u>** rent is used as a common denominator to compare leases with different provisions. This rent is the lease base rent less rent concessions.
25. The formula to calculate an operating expense ratio (OER) is: **<u>Total Expenses divided by Effective Gross Income</u>**.

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Chapter 3 | Development of Capitalization Rates

Exercise 3-1: Development of an R_o - Derivation from Comparable Sales - Solution

Sale No.	Sale Price	Remaining Economic Life	Land-to-Improvement Ratio	Expense Ratio	Net Operating Income	Overall Cap Rate
1	$760,000	18	1:4	34%	$64,600	8.5%
2	780,000	20	1:5	27%	66,300	8.5%
3	720,000	20	1:4	30%	64,800	9.0%
4	712,000	20	1:4	30%	64,000	9.0%
5	744,000	20	1:4	30%	67,000	9.0%
6	750,000	22	1:5	30%	82,500	11.0%

The six comparables and the subject property are of similar construction and condition. However, sales #3, #4, and #5, are most comparable to the subject. They have the same remaining economic life, land-to-improvement ratio, and income-to-expense ratio. The three sales indicate that an overall rate of .09 would be appropriate to use in valuing the subject property.

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Exercise 3-2: Development of an R_o - Derivation from Comparable Sales - Solution

	Comparable Sales		
	No. 1	**No. 2**	**No. 3**
Price	$2,100,000	$2,025,000	$2,550,000
No. of Units	25	24	30
Potential Gross Income	$410,000	$360,000	$450,000
V&C Loss	$18,450	$16,200	$20,250
Effective Gross Income	$391,550	$343,800	$429,750
Operating Expenses	$187,944	$151,272	$171,900
Net Operating Income	$203,606	$192,528	$257,850
Overall Capitalization Rate	9.7%	**9.5%**	10.1%
Remaining Economic Life	25 years	**25 years**	25 years
Operating Expense Ratio	48%	**44%**	40%
Land Value	$420,000	$405,000	$637,500
Land Percentage of Value	20%	**20%**	25%

From the comparable sales shown above the appropriate overall capitalization rate to use in your appraisal of a 25-unit apartment complex would be 9.5%. All comparable sales are quite similar to the subject; however comparable sale No. 2 is most similar to the subject in operating expense ratio and in land-to-building ratio. Thus, it is appropriate to use 9.5% in your appraisal of the subject property.

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Exercise 3-3: Development of an R_o - Derivation from Comparable Sales - Solution

Reconstructed Operating Statement	
Gross Income	**$3,450,245**
Vacancy & Collection loss	- 172,512
Effective gross income	**$3,277,733**
Expenses:	
Maintenance	393,328
Water & Sewer	65,700
Pool Cleaning	15,500
Insurance	115,000
Electricity	491,500
Natural Gas	398,000
Management Expense	196,000
Total Expenses	**$1,675,028**
Net Operating Income	**$1,602,705**
Overall Capitalization Rate (R_O)	**8.9%**

After properly reconstructing the owner's operating statement, the appraiser can develop the overall capitalization rate (R_O) from this market transaction.

The overall capitalization rate (R_O) can be computed using the IRV formula by dividing the net operating income of $1,602,705 by the sale price of $18,000,000 resulting in an overall capitalization rate (R_O) of 8.9%.

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Exercise 3-4: Development of an R_O - Band of Investment Technique Using Mortgage & Equity Components - Solution

You have determined that typical properties, like the subject you are appraising, are financed with 75% debt and 25% equity. From discussions with lenders and investors you have found that typical financing terms for this type of property require 9.3% of the amount borrowed be paid annually to amortize the loan, and the typical equity dividend rate (or equity capitalization rate – R_E) would be 10% for the equity investor. The effective tax rate is 2.5%.

From this information develop the overall unloaded capitalization rate (R_O) applicable to the subject property.

Capital	Portion		Cap Rate		Product
Debt	0.75	×	0.093	=	0.06975
Equity	0.25	×	0.100	=	0.02500
Totals	1.00		**Overall Rate (R_O) :**		**0.09475**

Overall Unloaded Capitalization Rate: **0.09475 or 9.5%**

Typical information from lenders and investors provided the necessary data to compute a capitalization rate for the subject property of 0.09475 or 9.5% rounded to 3 decimal places. The capitalization rate (R_O) for the subject property is a weighted average of the mortgage capitalization rate (R_M) (or annual constant) and the equity capitalization rate (R_E).

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Exercise 3-5: Development of an R_o - Band of Investment Technique Using Mortgage & Equity Components - Solution

You have been asked to develop an overall capitalization rate (R_O) for use in a commercial appraisal. You believe that similar commercial properties are financed with 80% debt and 20% equity. From discussions with investors and lenders, you determined that investors require an equity capitalization rate (R_E) of 6% for properties such as the subject and lenders require an 8% yield on loans of this type of mortgage, financed for 25 years with monthly payments. From this information you know that the mortgage capitalization rate (R_M) is 0.092616. The effective tax rate for the assessment jurisdiction is 2%. [*You determined this mortgage capitalization rate (RM) by looking in the compound interest tables or using a financial calculator – discussed in the next Chapter*]

From this information develop the overall unloaded capitalization rate (RO) applicable to the subject commercial property.

Capital	Portion		Cap Rate		Product
Debt	0.80	x	0.092616	=	0.0741
Equity	0.20	x	0.060000	=	0.0120
Totals	1.00		**Overall Rate (R_O)**	=	**0.086**

Based on the information provided from your discussions with lenders and investors, you were able to obtain the appropriate mortgage capitalization rate (R_M) of 9.2616% (which is discussed further in Chapter 4) and the equity capitalization rate (R_E).

Thus, the overall unloaded capitalization rate (R_O) can be computed using the band of investment technique. The indicated overall capitalization rate (R_O) for the subject property is a weighted average of the mortgage capitalization rate (R_M) (or annual constant) and the equity capitalization rate (R_E) or 0.086 (8.6%) rounded to three decimals.

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Exercise 3-6: Development of an R_O - Use of the Net Income Ratio Method - Solution

1. What is the potential gross income?

   10 apts. x 12 mos. x $1,500/mo = **$180,000**

2. What is the effective gross income (EGI)?

Potential Gross Income	$180,000
Less Vacancy & Collection Loss @ 6%	- 10,800
Effective Gross Income (EGI)	**$169,200**

3. What is the net operating income?

Effective Gross Income	$169,200
Less Operating Expenses	- 53,721
Net Operating Income	**$115,479**

4. What is the operating expense ratio (OER)?

   Operating expenses divided by effective gross income = OER

   $53,721 ÷ $169,200 = **31.75%**

5. What is the net income ratio?

   1 - OER or net operating income divided by effective gross income

   100% - 31.75% = 68.25% or $115,479 ÷ $169,200 = **68.25%**

6. What is the effective gross income multiplier?

> EGIM = sale price divided by effective gross income
>
> $1,100,000 ÷ $169,200 = **6.5 (EGIM)**

7. What is the overall capitalization rate?

> R_O = Net income ratio ÷ gross income multiplier
>
> 68.25% ÷ 6.5 = **10.5% overall capitalization rate**

Proof by Market Comparison Method using IRV formula:

> Net operating income of $115,479 ÷ sale price of $1,100,000 = **10.5%**

Exercise 3-7: Development of an R_o - Use of the Net Income Ratio Method - Solution

Develop the overall capitalization rate (R_O) from the following information which you have developed from sales comparables in the area of your subject property:

Potential gross income	=	$1,545,000
Vacancy and collection loss rate	=	5%
Operating expenses	=	$543,068
Effective gross income multiplier	=	7.0

Formula: R_O = (1 – OER) ÷ Effective gross income multiplier (EGIM)

Step 1: Calculate the effective gross income (EGI)

Formula: EGI = PGI – V&C

Potential gross income (PGI)	1,545,000
Vacancy & collection loss (5%)	77,250
Effective gross income (EGI)	1,467,750

Step 2: Calculate the operating expense ratio (OER)

Formula: OER = Total operating expenses ÷ EGI

OER = $543,068 ÷ $1,467,750 = **0.37**

Step 3: Calculate the net income ratio (NIR)

Formula: NIR = 1 – OER

NIR = 1 – 0.37 = **0.63**

Step 4: Calculate the overall capitalization rate (R_O)

Formula: RO = NIR ÷ EGIM

R_O = 0.63 ÷ 7.0 = **0.09 or 9.0%**

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Exercise 3-8: Development of an R_o - Use of the Debt Coverage Ratio Method - Solution

You have determined that a commercial property can expect a net operating income of $930,000 from an analysis of market information. Further, it is expected that the annual debt service for the property will be $620,000. You have determined that 75% of the property's value can be financed by debt with an annual mortgage requirement of 8% of the amount financed.

What is the overall capitalization rate applicable to this property? The debt coverage ratio (DCR) is calculated as follows:

Formula: **DCR** = NOI ÷ IM

DCR = $930,000 ÷ $620,000 = 1.50

If a mortgage can be obtained to finance 75% of the value of the property with an annual mortgage requirement of 8% of the amount financed, the overall capitalization rate is calculated as follows:

Formula: $\mathbf{R_O} = DCR \times R_M \times M$

$\mathbf{R_O} = 1.50 \times 0.08 \times 0.75$

$\mathbf{R_O = 0.09 \text{ or } 9.0\%}$

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Exercise 3-9: Development of an Overall Capitalization Rate (R_o) by Use of the Debt Coverage Ratio Method - Solution

Based on research and market analysis, you have learned that mortgage lenders in your area require a debt coverage ratio of 1.4 for the particular type of commercial property you are asked to appraise. The lenders have noted they will typically finance 60% of the investment for a term of 20 years in these commercial ventures, and the lenders require a 10% yield on their loans, which require level monthly payments of principal and interest. By using a financial calculator, you have determined that a 10% loan for 20 years with level monthly payments indicates a mortgage annual constant (or mortgage capitalization rate – R_M) of 11.58%.

Based upon the above data, what would be the indicated overall capitalization rate (R_O) indicated for the subject commercial property?

The formula for the overall capitalization rate (R_O) by the DCR method is:

Formula: $R_O = DCR \times R_M \times M$

where:

DCR = Debt Coverage Ratio

R_M = Annual Mortgage Constant (or Mortgage Cap Rate)

M = Loan-to-Value Ratio

Formula: $R_O = DCR \times R_M \times M$

$R_O = 1.4 \times 0.1158 \times 0.60$

$\mathbf{R_O = 0.097}$ **or 9.7%**

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Exercise 3-10: Development of an Overall Capitalization Rate (R_o) by The Band of Investment Technique Using Land & Building Components - Solution

You are developing overall capitalization rates for retail stores as a part of your mass appraisal process. You have determined that typical retail developments similar to the stores you are appraising are made up of land and building components of 30% and 70%, respectively.

From market extraction techniques, you have learned that the land and building capitalization rates for similar properties are 8% and 12%, respectively.

Determine the overall capitalization rate using the band-of-investment method (weighted average of land and building capitalization rates).

The band-of-investment technique using the land capitalization rate (R_L) and building capitalization rate (R_B) provides the weighted average of these two rates and this result is also known as the overall capitalization rate (R_O).

Property Component	Percent of Investment		Cap Rate		Product
Land	0.30	X	0.08	=	0.024
Improvement	0.70	X	0.12	=	0.084
Totals	1.00	**Overall Rate (R_o)**		=	0.108

R_O = 0.108 or 10.8%

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Exercise 3-11: Development of an Overall Capitalization Rate (R_O) by Various Methods - Solution

A: Derivation from comparable sales:

Calculate the net operating income:

Effective gross income	$1,200,000
Less operating expenses (45%)	- 540,000
Net Operating Income	**$ 660,000**

Formula: R_O = NOI ÷ Sale Price

R_O = $660,000 ÷ $6,000,000 = 0.11 or **11%**

B: Debt coverage ratio method

Property sale price	$6,000,000
Loan-to-Value Ratio (M)	x .60
Amount of mortgage	**$3,600,000**
Annual Mortgage Constant R_M	x .10
Amount of annual debt service (I_M)	**360,000**

The debt coverage ratio is calculated as follows:

Formula: **DCR** = NOI ÷ I_M

DCR = $660,000 ÷ $360,000 = 1.8333

Formula: R_O = DCR x R_M x M

R_O = 1.8333 x 0.10 x 0.60

R_O = 0.11 or **11%**

(continued from previous page)

C: Net income ratio method:

The formula for determining the overall capitalization rate (RO) using the net income ratio method is:

$$R_O = \frac{NIR}{EGIM}$$

Where:
NIR = Net income ratio
EGIM = Effective gross income multiplier

The net income ratio is 0.55 which is computed by deducting the operating expense ratio of 45% from 100%. The gross income multiplier is found by dividing the sales price of $6,000,000 by the effective gross income of $1,200,000 which equals 5.00. With this information, the overall capitalization rate (R_O) would be:

$$R_O = \frac{0.55}{5.00} = 0.11 = \mathbf{11\%}$$

D: Band of investment technique (mortgage & equity cap rates)

Capital	Portion		Cap Rate		Product
Debt	0.60	x	0.1000	=	0.0600
Equity	0.40	x	0.1250	=	0.0500
Totals	1.00		**Overall Rate (R_O)**	=	**0.1100**

The weighted average of the debt and equity cap rates is 0.11 or **11%.**

Exercise 3-12: Extracting the Property's Overall Yield Rate (Y_O) by The Band of Investment Method - Solution

You have been asked to derive the overall yield rate (Y_O) for a public utility property that is expected to be financed with interest only debt requiring a balloon payment at maturity to pay off the mortgage. You believe this is typical financing for utility properties of this type, because most appear to be financed with long-term bonds. Lenders have informed you that their yield requirement for this type of debt is 8% and you know that equity investors require a 12% yield for this level of risk.

Market information indicates that investors in this type of property typically finance 40% of this type of purchase with debt and 60% with equity. Further, you believe there will be no change in value or income during the holding period because of regulatory restrictions.

What is the indicated overall yield rate (Y_O) for this investment?

You recognize you can use the band of investment method to derive the overall yield rate for the property in this instance because the public utility property is to be financed with an interest only loan and there is not expected to be any change in income or value during the investment holding period.

Financing	Portion		Yield Rate		Product
Debt	0.40	x	0.08	=	0.032
Equity	0.60	x	0.12	=	0.072
Totals	1.00		**Yield Rate (Y_O)**	=	**0.104**

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Exercise 3-13: How Tax Rates are Expressed - Solution

No.	Tax Rate	Decimal Equivalent
1	$65.00/$1,000	**0.065**
2	72 Mills	**0.072**
3	$8.00/$100	**0.080**
4	$80.00/$1,000	**0.080**
5	115 Mills	**0.115**
6	$4.50/$100	**0.045**
7	62.5 Mills	**0.0625**
8	$75.25/$1,000	**0.07525**
9	$12.75/$100	**0.1275**
10	$6.25/$100	**0.0625**

Table: How Tax Rates Are Calculated

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Exercise 3-14: Development of the Effective Tax Rate - Solution

1. Complete the following table of equivalencies:

Mills		$ Per $100		Nominal Tax Rate*
26.0	=	2.60	=	***0.0260***
35.0	=	***3.50***	=	***0.0350***
10.0	=	***1.00***	=	0.0100
72.5	=	7.25	=	0.0725
6.0	=	***0.60***	=	***0.0060***
15.0	=	***1.50***	=	0.0150

* Express as a decimal.

2. Complete the following table. Utilize the EAT triangle formula.

Assessment Level	Nominal Tax Rate $ Per $100	Mills	Nominal Tax Rate	Effective Tax Rate
0.25	$5.50	***55.0***	***0.055***	***0.01375***
0.60	***$6.00***	60.0	***0.060***	0.03600
0.35	***$8.50***	***85.0***	***0.085***	0.02975
0.50	***$6.25***	62.5	***0.0625***	***0.03125***
0.40	***$7.00***	70.0	***0.070***	0.02800
0.30	$7.75	***77.5***	***0.0775***	0.02325

3. Complete the following table:

Market Value	Tax Dollars	Effective Tax Rate*
$560,000	$12,880	***0.023***
$900,000	***$29,700***	0.033
$680,000	$19,040	***0.028***
$1,300,000	$32,500	0.025
$940,000	***$24,440***	0.026
$660,000	$23,100	0.035

* Express as a decimal.

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Exercise 3-15: Development of the Effective Tax Rate by Market Comparison - Solution

A. Market Comparison

(Tax bill) $5,400 ÷ $360,000 = 0.015 or **1.5%**

B. EAT formula

(Assessment Level) 0.50 x .030 (Tax Rate) = 0.015 or **1.5%**

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Exercise 3-16: Development of the Effective Tax Rate by Market Comparison - Solution

This problem can be solved using the IRV equation where the taxes paid are in the numerator and the Sale price is in the denominator.

Tax Jurisdiction North Side

Sale A: Effective tax rate = taxes paid ÷ sale price ETR = $7,560 ÷ $420,000 = 0.018 or **1.8%**

Sale B: Effective tax rate = taxes paid ÷ sale price ETR = $6,750 ÷ $375,000 = 0.018 or **1.8%**

Tax Jurisdiction South Side

Sale C: Effective tax rate = taxes paid ÷ sale price ETR = $2,063 ÷ $165,000 = 0.0125 or **1.25%**

Sale D: Effective tax rate = taxes paid ÷ sale price ETR = $2,300 ÷ $184,000 = 0.0125 or **1.25%**

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REVIEW QUESTIONS - Chapter 3 SOLUTION

1. Name five methods of developing an overall capitalization rate (R_O).

   - **Derivation from comparable sales**
   - **Band of investment (mortgage and equity components)**
   - **Net income ratio**
   - **Debt coverage ratio**
   - **Band of investment (land and improvement components)**

2. The summation method of estimating an overall yield rate (YO) [property discount rate] is a theoretical method which is based upon the sum of four components that affect the return on or requirements of an investment. The four components are:

   - **Safe rate**
   - **Risk rate**
   - **Non-liquidity rate**
   - **Management rate**

3. When developing the overall capitalization rate from market comparisons, three critical items which must be highly comparable between comparable sales and the subject property are:

   - **Land-to-improvement ratio**
   - **Operating expense ratio**
   - **Remaining economic life**

4. Another name for the mortgage capitalization rate (R_M) is the **annual mortgage constant**.

5. The **band of investment** technique computes a weighted average of the mortgage capitalization rate (R_M) and equity capitalization rate (R_E). The resulting weighted average is called the overall capitalization rate (R_O).

6. The capitalization rate method most often use by lending institutions is the **debt coverage ratio** method.

7. The **effective tax rate** is the percentage that annual real estate taxes are in relation to total property value.
8. Name the two methods of developing an effective tax rate.
   - **EAT formula**
   - **Market comparison**
9. The net income ratio method can be used to develop an overall capitalization rate when the net income ratio is divided by the **effective gross income multiplier**.
10. What are the three components of an overall capitalization rate:
    - **Overall yield rate**
    - **Recapture rate**
    - **Effective tax rate**
11. The **net income ratio method** for developing an overall capitalization rate is appropriate when the effective gross income multiplier is known.
12. The **land capitalization rate (R_L)** reflects the relationship between the land income and land value.
13. The weighted average of the land capitalization rate and the building capitalization rate is known as the **overall capitalization rate**.
14. Dividing the building (improvement) income by the **building (improvement) capitalization rate** will result in the building (improvement) value.
15. The assessment level is 30% of appraised value and the current tax rate is $8.50 per hundred. What is the effective tax rate expressed as a decimal and percentage? **0.0255 or 2.55% (rounded)**

Chapter 4 | Contemporary Capitalization Methods

Exercise 4-1: Direct Capitalization of Land- Solution

	Sale 1	**Sale 2**	**Sale 3**
Sale Price	$1,950,000	$1,800,000	$2,400,000
Square Foot Area	31,850	33,000	46,500
Shape	Irregular	Irregular	Rectangular
Topography	Level	Level	Level
Operating Expenses Ratio (OER) *Excludes* Taxes	21%	19%	15%
Net Operating Income (NOI) *Includes* Taxes	$126,000	$118,000	$143,000
Overall Rate (R_O) Unloaded (NOI ÷ Sale Price)	**0.0646**	**0.0656**	**0.0596**

In selecting the overall rate for each subject, it is necessary to consider the similarities and differences between the subjects and the comparables. Subject A has the most similarities with Sales 1 and 2, while subject B is most similar to Sale 3.

Considering these similarities, the appropriate unloaded R_O for Subject A would be 0.065, and the loaded R_O would be 0.085. The appropriate unloaded R_O for Subject B would be 0.060, and the loaded R_O would be 0.080.

The indicated value of Subject A would be:

$161,500 ÷ 0.085 = **1,900,000**.

The indicated value of Subject B would be:

$183,600 ÷ 0.080 = **$2,295,000**.

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Exercise 4-2: Direct Capitalization of Improved Property- Solution

Effective tax rate for subject: 30 mills ÷ 1,000 = 0.03 × 0.50 assessment level = **0.015**

Income & expense statement for the subject and comparables:

	Sale 1	Sale 2	Sale 3	Subject
Potential Gross Income	$900,000	$980,400	$1,121,000	$940,000
V&C Loss (5%)	– 45,000	– 49,020	– 56,050	– 47,000
Miscellaneous Income	+ 7,800	+ 8,220	+ 10,400	+ 7,000
Effective Gross Income	862,800	939,600	1,075,350	900,000
Operating Expenses & Reserves	(30%) –258,840	(30%) –281,880	(34%) –365,619	(30%) –270,000
NET OPERATING INCOME	**$603,960**	**$657,720**	**$709,731**	**$630,000**

Calculations for tax expenses and overall capitalization rates (unloaded) for comparables:

Sale 1:
$7,200,000 sale price × 0.0190 ETR = $136,800 tax expense
603,960 NOI – 136,800 = 467,160 NOI *including* taxes ÷ 7,200,000 **= 0.0649 R_O**

Sale 2:
$7,600,000 sale price × 0.0160 ETR = $121,600 tax expense
657,720 NOI – 121,600 = 536,120 NOI *including* taxes ÷ 7,600,000 **= 0.0705 R_O**

Sale 3:
$8,000,000 sale price × 0.0175 ETR = $140,000 tax expense
709,731 NOI – 140,000 = 569,731 NOI *including* taxes ÷ 8,000,000 **= 0.0712 R_O**

Sale #1 is the most comparable to the subject property overall, and the rates indicated by Sales 2 and 3, while higher, do support the rate as indicated by Sale 1. Therefore, a rate of 0.065 (rounded) is selected as the overall rate for the subject without a tax component (i.e., “unloaded”). Adding the effective tax rate for the subject's jurisdiction (0.015) provides an overall rate (“loaded”) of 0.080.

The value of the subject: **$630,000 ÷ 0.08 = $7,875,000**

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Exercise 4-3: Direct Capitalization with an Overall Rate by the Band of Investment Technique- Solution

Gross Square Feet	80,000
Common Area	12%
Effective Tax Rate	0.014
Market Rent ($18.00 x 70,400 SQFT)	$1,267,200
Vacancy Rate (10%)	-126,720
Effective Gross Income	$1,140,480
Expenses (40%)	-456,192
Net Operating Income	$684,288

Overall Rate:
The monthly partial payment factor of 0.009087 (10% loan, for 25 years) is multiplied by 12 to obtain the mortgage annual constant or the mortgage capitalization rate (0.009087 x 12 = 0.109044).

Financial Components	Percent of Investment		Rate		Product
Debt	0.70	x	0.109044	=	0.076331
Equity	0.30	x	0.15	=	0.045000
Totals	1.00			**Overall Rate (R_O) =**	0.121331

Overall Capitalization Rate (R_O) without a tax component (rounded) = 0.121. Adding in the effective tax rate (0.014) provides an overall capitalization rate of 0.135.

The indicated value of the subject: **$684,288 ÷0.135 = $5,068,800**

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Exercise 4-4: Direct Capitalization with an Overall Rate by Net Income Ratio - Solution

The net income ratio can be obtained by subtracting the expense ratio from 1.00. The subject property has expenses of 35% or an expense ratio of 0.35. Subtracting the expense ratio from 1.00 provides a net income ratio of 0.65.

The net income ratio can also be obtained by dividing the net operating income by the effective gross income. The subject has a potential gross income of $260,000, less $13,000 for vacancy and collection loss, providing an effective gross income (EGI) of $247,000. Subtracting expenses, which are 35% of effective gross income, provides a net operating income (NOI) of $160,550. Dividing the NOI $160,550 by the EGI $247,000 provides a net income ratio of 0.65.

The overall capitalization rate is calculated as follows:
.65 ÷5 = 0.13

Indicated value of subject property: NOI ÷ R_O = Value

$160,550 ÷ 0.13 = $1,235,000

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Exercise 4-5: Direct Capitalization with an Overall Rate by the Debt Coverage Ratio - Solution

A property comparable to your subject property has a net operating income of $600,000 with annual debt service of $400,000. The property was financed with a loan for 70% of its value and a mortgage constant of 11.58%.

In order to develop an overall capitalization rate, the Debt Coverage Ratio must be calculated. It is calculated as follows:

$600,000 ÷ $400,000 = 1.50

The overall rate can now be calculated by

the formula: $R_O = DCR \times R_M \times M$

The calculation for the overall capitalization rate

would be: 1.50 x 0.1158 x 0.70 = 0.12159

rounded to 12.2%.

The indicated value for the subject property would be:

$420,000 ÷ 0.122 = $3,442,622

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Exercise 4-6: Development of Gross Income (Rent) Multipliers - Solution

A retail store containing 16,000 square feet of gross leaseable area is rented for $16.00 per square foot per year. Vacancy and collection loss is 4%. Allowable expenses are 40% of effective gross income. The land value is estimated to be $384,000. The store sold recently for $1,536,000.

The calculation for the Gross Income Multiplier (GIM) is:

16,000 s.f. × $16.00 = $256,000 PGI

$1,536,000 ÷ $256,000 = **6.00** (GIM)

The calculation for the Effective Gross Income Multiplier (EGIM) is:

$256,000 – Vacancy & Collection loss (4%) = $245,760

$1,536,000 ÷ $245,760 = **6.25**

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Exercise 4-7: Development of an Annualized Mortgage Constant - Solution

You are performing a reappraisal on all the offices in the central business district. You have decided to use the mortgage & equity band of investment technique. You have already calculated an equity capitalization rate of 12% and now need to calculate the mortgage constant. Lending institutions require a loan to value ratio of 70% for office properties and the typical terms of the loan for these offices are twenty years with monthly payments with an interest rate of 8%. What is the annual mortgage constant (R_M)?

Solution using the Compound Interest Table:

1. Find the 8% monthly compound interest table.
2. Using column 6 (Installment to Amortize) find the factor for 20 Years (240 Mths)

    Factor = 0.008364

3. Multiply the monthly factor by 12 to annualize the rate.

    R_M = 0.008364 × 12 = 0.100368

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REVIEW QUESTIONS - Chapter 4 SOLUTION

1. **Direct capitalization** is a method of converting an estimate of a single year's income into value in one direct step.
2. The two generic formulas used in direct capitalization are:
   - **IRV, which is income divided by rate equals value**
   - **VIF, which is income times a factor equals value**
3. **Yield capitalization** is a method of converting future net benefits into present value where each future net benefit is discounted at a proper yield rate (discount rate).
4. The **Partial Payment Factor** shows the periodic payment necessary to amortize a loan at a specified interest rate over a specific number of periods.
5. The **Present Worth of $1 per Period** shows the present worth of a series of future payments or deposits.
6. The **Present Worth of $1** shows the present worth of a single future payment or deposit of $1.
7. Write the formula for developing a gross income multiplier.
   GIM = Sales Price ÷ Gross Income
8. List three comparability criteria necessary when utilizing direct capitalization.
   - **Land-to-improvement (building) ratios**
   - **Expense ratios**
   - **Remaining economic life (REL)**
9. A comparable sale property to the subject has a net operating income of $150,000 inclusive of taking real estate taxes as an expense. The property sold for $1,500,000 and is located in the subject's tax jurisdiction, which has a 1% effective tax rate. The subject property has an expected net operating income of $132,000 exclusive of taking real estate taxes as an expense.

   What is the subject property's value? **$1,200,000**

   $150,000 ÷ $1,500,000 = 0.10

   0.10 + 0.01 = 0.11

   $132,000 ÷ 0.11 = **$1,200,000**

10. A property recently sold for $2,000,000. At the time of sale, the property had a potential gross income of $270,000, and an expected vacancy loss of $20,000.

    What is the effective gross income multiplier? **<u>8.00</u>**

    $2,000,000 ÷ (270,000 – 20,000) = **8.00**

11. When property is being appraised for ad valorem tax purposes the **<u>effective tax rate</u>** must be included as a part of the overall capitalization rate.

12. List four types of amenities that must be considered when selecting vacant land sales.

    - **Location**
    - **Restrictions**
    - **Investment (economic) desirability**
    - **Utility**

13. List the four major categories of required comparability that must be considered when selecting improved property sales to use in the development of an overall capitalization rate.

    - **Amenities**
    - **Land-to-improvement ratio**
    - **Expense ratios**
    - **Remaining economic life**

14. **<u>Direct capitalization</u>** is different from yield capitalization, in that it does not directly consider the individual cash flows beyond the first year.

15. The formula value equals **<u>income ÷ rate</u>** is one of the generic formulas used in direct capitalization.

Chapter 5 | Historical Capitalization Methods

Exercise 5-1: Developing a Recapture Rate by the Market Comparison Technique- Solution

Net Operating Income:	$240,000
Discount Dollars ($1,800,000 x 0.085):	- 153,000
Tax Dollars ($1,800,000 x 0.020):	- 36,000
Net Income After Deduction for Discount and Taxes:	$51,000
Recapture Rate:	
Recapture Income:	$51,000
Recapture Rate ($51,000 ÷ $1,200,000):	**0.0425**

Note: The years of remaining economic life can be obtained by dividing the recapture rate into 1.

1 ÷ 0.0425 = 23.5 years of remaining economic life.

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Exercise 5-2: Developing a Land Capitalization Rate by the Market Comparison & Summation Methods- Solution

A. Summation

Overall Yield Rate	0.085
Effective Tax Rate (0.50 assessment level × 0.04 tax rate)	+0.020
Land Capitalization Rate	**0.105**

B. IRV Equation (market comparison)

\$52,500 (Land income) ÷ \$500,000 (land value) = 0.105 Land Cap Rate (R_L)

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Exercise 5-3: Developing a Building Capitalization Rate by the Market Comparison & Summation Methods- Solution

A. Summation

Overall Yield Rate	0.10
Effective Tax Rate (0.25* x 0.08)	0.02
Recapture Rate (1 ÷ 25)	+0.04
Improvement Capitalization Rate	**0.16**

B. IRV Equation

Net Operating Income	$171,000
Less Income to Land (0.12** x $225,000)	-27,000
Improvement Income =	**$144,000**

(Improvement Income) $144,000 ÷ $900,000 (Improvement Value) = **0.16**

*__Assessment Level__ is determined by dividing the assessed value by the sale price.

The **Land Capitalization Rate (R_L) is the sum of the overall yield rate and the effective tax rate.

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Exercise 5-4: Straight Line Capitalization - Land Residual Technique - Solution

Effective gross income..$225,000

Less expenses

Insurance $2,400
Management (4%)...9,000
Snow removal, etc..18,000
Legal and accounting (1%) 2,250
Miscellaneous 3,600
Total Expenses ....– 35,250

NOI before discount, recapture and taxes $189,750

Less income to improvement
(9.5% + 10.0% + 2.5%) = 22.0%
$300,000 x 0.22 ....–$66,000

Residual income to land..$123,750
Capitalized @ 12% (9.5% + 2.5%)....................................$1,031,250
Plus improvement value...$300,000
Total property value $1,331,250

ALTERNATE SOLUTION:
NOI before discount, recapture and taxes $189,750
Less annual paving reserve for replacement ....–30,000
Residual income to property $159,750
Capitalized @ 12% (9.5% + 2.5%) $1,331,250

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ALTERNATE SOLUTION (SOLVE USING MATRIX):

Step 1: Insert building value and net operating income

	I	R	V
Bldg			**$300,000**
Land			
Total	**$189,750**	–	

Step 2: Insert building and land capitalization rates

	I	R	V
Bldg		**0.220**	$300,000
Land		**0.120**	
Total	$189,750	–	

Step 3: Multiply (right to left) across the "building" row

	I	R	V
Bldg	**$66,000**	0.220	$300,000
Land		0.120	
Total	$300,000	–	

Step 4: Subtract income attributable to building from total income to find residual income attributable to land

	I	R	V
Bldg	$66,000	0.220	$300,000
Land	**$123,750**	0.120	
Total	$189,750	–	

Step 5: Divide (left to right) across "land" row

	I	R	V
Bldg	$66,000	0.220	$300,000
Land	$123,750	0.120	**$1,031,250**
Total	$189,750	–	

Step 6: Add building value and land value to find total property value

	I	R	V
Bldg	$66,000	0.220	$300,000
Land	$123,750	0.120	$1,031,250
Total	$189,750	–	**$1,331,250**

Exercise 5-5: Straight Line Capitalization - Land Residual Technique - Solution

Potential Gross Income (12,500 sf @ $35.00)		$ 437,500
Less Vacancy and Collection Loss (2%)		-8,750
Effective Gross Income		$ 428,750
Less Expenses:		
Insurance	$ 7,700	
Maintenance and Repair	15,750	
Legal & Accounting	4,200	
Utilities	12,600	
Miscellaneous	5,600	
Management (428,750 x 5%)	21,438	
Trash Service	2,100	
Replacement Reserves	7,000	
Total Expenses		- 76,388
Net Operating Income		$ 352,362
Less Income Attributable to the Building (12,500 sq. ft. @ $140.00 = $1,750,000 x 0.16) =		- 280,000
Residual Income Attributable to the Land		$ 72,362
Capitalized at 0.135		$536,015
		Rounded to **$536,000**

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ALTERNATE SOLUTION (SOLVE USING MATRIX):

Step 1: Insert building value and net operating income

	I	R	V
Bldg			**$1,750,000**
Land			
Total	**$352,362**	–	

Step 2: Insert building and land capitalization rates

	I	R	V
Bldg		**0.160**	$1,750,000
Land		**0.135**	
Total	$352,362	–	

Step 3: Multiply (right to left) across the "building" row

	I	R	V
Bldg	**$280,000**	0.160	$1,750,000
Land		0.135	
Total	$352,362	–	

Step 4: Subtract income attributable to building from total income to find residual income attributable to land

	I	R	V
Bldg	$280,000	0.160	$1,750,000
Land	**$72,362**	0.135	
Total	$352,362	–	

Step 5: Divide (left to right) across "land" row to find land value

	I	R	V
Bldg	$280,000	0.160	$1,750,000
Land	$72,362	0.135	**$536,015**
Total	$352,362	–	

Exercise 5-6: Straight Line Capitalization - Building Residual Technique - Solution

Potential gross income (20 x 1,200 x $25) =		$ 600,000
Less: Vacancy & collection loss (5%)		- 30,000
Effective gross income		$570,000
Total operating expenses		- 171,000
Net operating income		$399,000
Less Income attributable to land:		
($600,000 x 0.115) =		- 69,000
Net income to building =		$330,000
Building capitalization rate:		
Overall yield rate	0.090	
Recapture rate (1÷25)	0.040	
Effective tax rate	0.025	
Building cap rate =	0.155	
Income to building capitalized at 0.155 =		$ 2,129,032
Add land value		+ 600,000
Total indicated value of improved property		**$2,729,032**
	Rounded to	**$2,729,000**

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ALTERNATE SOLUTION (SOLVE USING MATRIX):

Step 1: Insert land value and net operating income

	I	R	V
Bldg			
Land			**$600,000**
Total	**$399,000**	–	

Step 2: Insert building and land capitalization rates

	I	R	V
Bldg		**0.155**	
Land		**0.115**	$600,000
Total	$399,000	–	

Step 3: Multiply (right to left) across the "land" row

	I	R	V
Bldg		0.155	
Land	**$69,000**	0.115	$600,000
Total	$399,000	–	

Step 4: Subtract income attributable to land from total income to find residual income attributable to building

	I	R	V
Bldg	**$330,000**	0.155	
Land	$69,000	0.115	$600,000
Total	$399,000	–	

Step 5: Divide (left to right) across "building" row to find building value

	I	R	V
Bldg	$330,000	0.155	**$2,129,032**
Land	$69,000	0.115	$600,000
Total	$399,000	–	

Step 6: Add building value and land value to find total property value

	I	R	V
Bldg	$330,000	0.155	$2,129,032
Land	$69,000	0.115	$600,000
Total	$399,000	–	**$2,729,032**

Exercise 5-7: Straight Line Capitalization - Building Residual Technique - Solution

Potential gross income			$ 2,354,000
Less: Vacancy & collection loss (4%)			- 94,160
Effective gross income			$2,259,840
Total operating expenses			- 660,000
Net operating income			$1,599,840
Less income attributable to land: ($2,500,000 x 0.10) =			- 250,000
Net income to building =			$1,349,840
Building capitalization rate:			
Overall yield rate	0	.08	
Recapture rate (1÷20)	0	.05	
Effective tax rate	0	.02	
Building cap rate =	0	.15	
Income to building capitalized at 0.15 =			$ 8,998,933
Add land value			+ 2,500,000
Total indicated value of property			$11,498,933
			Rounded **$11,500,000**

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ALTERNATE SOLUTION (SOLVE USING MATRIX):

Step 1: Insert land value and net operating income

	I	R	V
Bldg			
Land			**$2,500,000**
Total	**$1,599,840**	–	

Step 2: Insert building and land capitalization rates

	I	R	V
Bldg		**0.150**	
Land		**0.100**	$2,500,000
Total	$1,599,840	–	

Step 3: Multiply (right to left) across the "land" row

	I	R	V
Bldg		0.150	
Land	**$250,000**	0.100	$2,500,000
Total	$1,599,840	–	

Step 4: Subtract income attributable to land from total income to find residual income attributable to building

	I	R	V
Bldg	**$1,349,840**	0.150	
Land	$250,000	0.100	$2,500,000
Total	$1,599,840	–	

Step 5: Divide (left to right) across "building" row to find building value

	I	R	V
Bldg	$1,349,840	0.150	**$8,998,933**
Land	$250,000	0.100	$2,500,000
Total	$1,599,840	–	

Step 6: Add building value and land value to find total property value

	I	R	V
Bldg	$1,349,840	0.150	$8,998,933
Land	$250,000	0.100	$2,500,000
Total	$1,599,840	–	**$11,498,933**

$11,500,000 rounded

REVIEW QUESTIONS - Chapter 5 SOLUTION

1. If the land capitalization rate in straight line capitalization (including the effective tax rate) is 9%, and the remaining economic life of the improvement is 25 years, the building capitalization rate is **13%.**

2. In straight-line capitalization, **recapture** is received in equal amounts during the economic life of the improvement.

3. Straight-line capitalization assumes a **declining** income stream during the remaining economic life of the improvements.

4. Straight-line capitalization is best utilized when the property has:
   - **Income is expected to decline over the remaining economic life**
   - **Discount (or overall yield) is earned on nondepreciated balance of investment**
   - **Recapture is received in equal amounts during remaining life of improvements**

5. The building residual technique should be used when:
   - **Land value supported by market data is known**
   - **Improvement is not new**
   - **Improvement is not adequate for site**
   - **Vacant land sales are available**

6. The land residual technique should be used when:
   - **Improvements are adequate in condition and suitability to the site**
   - **Land consists of a large parcel with no comparable property available**
   - **Land sales are not available**
   - **Building is new, and the proper use and cost are known**

7. The land capitalization rate is composed of what components?
   - **Discount rate (or overall yield rate)**
   - **Effective tax rate**

8. The building capitalization rate is composed of what components?
   - **Discount rate (or overall yield rate)**
   - **Recapture rate**
   - **Effective tax rate**
9. The overall yield rate (Y_O) is another name for the property's **discount** rate.
10. To obtain the land income in the building residual method one **multiplies** the land capitalization rate by the **land** value.
11. To obtain the **building** capitalization rate, one can divide the building income by the building **value.**
12. The building residual technique requires the development of the discount rate, recapture rate, effective tax rate, and a supportable market value for land.
    a. Land
    b. **Building**
    c. Both A and B
    d. None of the above

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